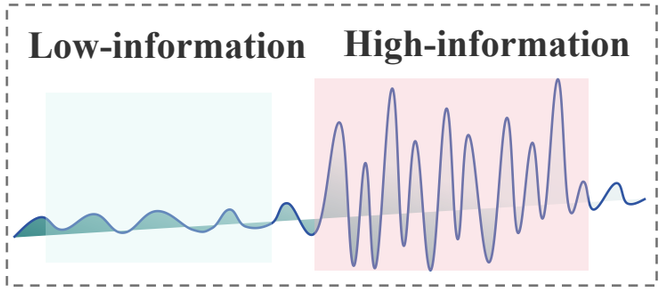
(a) Uniform Computation Allocation

Input Time Series $\mathbf{X}_{1:L}$



Uniform Computation Allocation



Same Complexity for All Tokens (t, s, h)

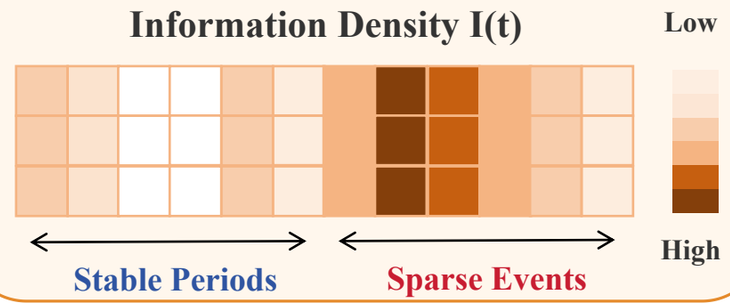
Wasted Computation on Simple Patterns

Insufficient Modeling for Critical Dynamics

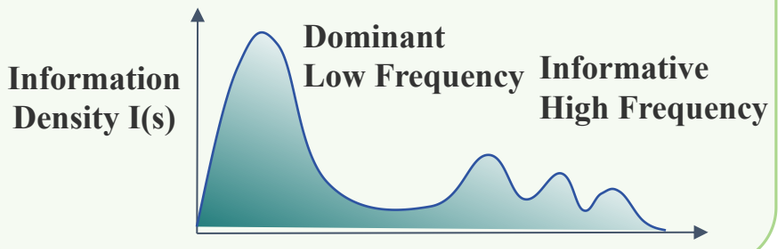
(b) Information Density Across Axes

Information Density Varies Across Three Axes

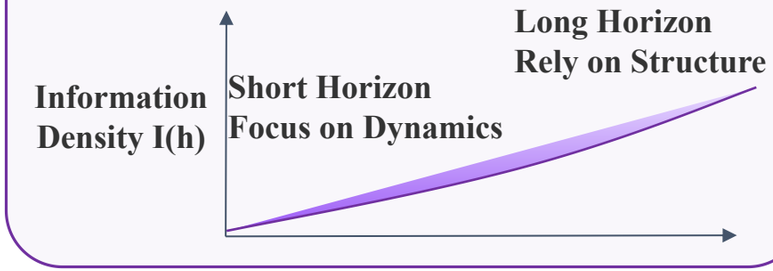
Temporal Dimension t



Spectral Dimension s



Horizon Dimension h



(c) BucpTSF: Adaptive Computation

Reallocate Computation to Match Information

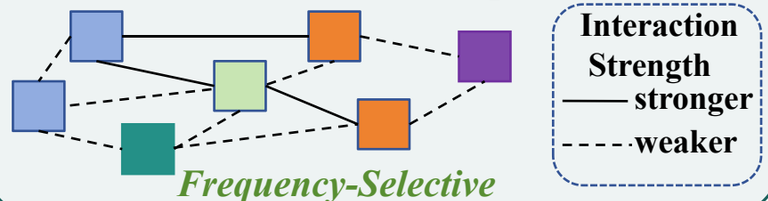
1) Representation Reorganization

Construct Heterogeneous Multi-level Tokens



2) Structured Token Interaction

Model More Informative Dependencies



3) Horizon-Conditioned Extrapolation

Adapt Prediction Complexity to Horizon



Principle: Allocate computation according to information density. $C(t,s,h) \propto I(t,s,h)$